Cyberly: A Gamified Android Application for Learning Basic Cybersecurity Concepts*

R. Muhammadi[1], P. Kokkinos[1,2], A. Kretsis[2,3], and E. Varvarigos[2,3]

[1] Department of Digital Systems, University of the Peloponnese, Sparta, Greece
ds19113@office365.uop.gr
[2] Institute of Communication and Computer Systems, Athens, Greece
kokkinop@mail.ntua.gr
[3] School of Electrical and Computer Engineering, NTUA, Athens, Greece
{akretsis, vmanos}@mail.ntua.gr

Abstract

This paper presents the design and development of Cyberly, a gamified Android application for learning basic cybersecurity concepts. The application is addressed to vulnerable Internet users such as children and young people, elderly individuals, and people with limited knowledge regarding digital security. Many of these users are daily Internet users without being aware of the risks it may involve and the threats they may face. The selection of this topic was based on the rapid development of digitalization and technology, especially in the current era with the use of Artificial Intelligence, which creates new challenges as well as security gaps. The work presents basic concepts of cybersecurity and analyzes common and dangerous Internet threats, while also using gamification in education in order to create a more understandable and engaging way of learning.

1 Introduction

The continuous spread of technology and digital systems, together with the increasing dependence of society on the Internet, has made cybersecurity one of the most important fields of our time. Daily activities such as communication, financial transactions, work, entertainment, and education are now carried out through digital systems, which significantly increases the exposure of users and organizations to Internet risks. Consequently, cybersecurity is a basic element for ensuring the smooth and secure operation of the digital environment.

Cybersecurity refers to the set of practices, technologies, and processes that are applied with the purpose of protecting digital systems, networks, data, and applications from malicious actions. These actions may include unauthorized access, alteration or deletion of data, and the theft of sensitive information. Cybersecurity is not limited only to data protection from a technical point of view, but also extends to risk management, prevention, and preparedness for the handling of security incidents. At the same time, the development of technology does not bring only positive changes; it also brings security gaps, the evolution of threats, and new forms of automated deception, especially in the era of artificial intelligence [1].

For this reason, cybersecurity is not only a technical matter but also a matter of education and culture. User awareness and education regarding safe Internet practices are essential for reducing risks. This paper presents the design and development of Cyberly, a gamified Android application that supports prevention through accessible learning.

*Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. 47, July 8–10, 2026, Seoul, South Korea.
© The copyright of this paper remains with the author(s).

Several gamified cybersecurity learning platforms have been proposed in the literature. In [2], the Cyber-Aware mobile application was designed to improve cybersecurity awareness among K–6 children by combining educational principles with both formal and informal learning activities. Similarly, authors in [3] developed a gamified Android role-playing quiz application that promotes password security awareness through interactive gameplay. [4] introduced an online gamified cybersecurity learning platform that integrates a custom game environment, automated feedback mechanisms, and personalized exercises to teach practical cybersecurity skills. Furthermore, [5] presented a game-based cybersecurity training platform featuring virtual laboratories, educational challenges, and motivational design elements. Collectively, these studies demonstrate the growing interest in applying gamification to cybersecurity education. Cyberly extends this direction by targeting vulnerable and non-technical users through a mobile-first learning experience that combines cybersecurity awareness content with interactive simulations and simplified gamification mechanisms.

The main contributions of this work are:

- The design of a gamified mobile application for cybersecurity education
- The integration of interactive simulations for experiential learning
- A lightweight MVP architecture that prioritizes accessibility and usability

The remainder of this paper is organized as follows. Section 2 provides the cybersecurity background and motivation, while Section 3 reviews related work and gamification in cybersecurity education. Sections 4–7 present the design, implementation, and evaluation of Cyberly. Finally, Section 8 concludes the paper and outlines future research directions.

2 Cybersecurity Background and the Need for Education

The continuous evolution of technology and digitalization, together with the extensive use of the Internet, has led to a complex threat environment. Modern Internet threats are no longer limited to simple attacks, but combine technology and human factors, making cybersecurity a highly demanding field. While more and more data are digitized, transmitted, and stored, the effects of a successful cyberattack are serious for individual users, organizations, and even entire states.

A main category of modern Internet threats is malicious software, or malware. Malware consists of programs designed to cause damage, steal data, or obtain unauthorized access to systems. Such threats may appear as viruses, trojans, spyware, or ransomware and are usually spread through infected files, links, and applications. Ransomware in particular remains one of the most dangerous attacks because it locks the user data and demands money for its restoration [7].

Another major category of modern threats is social engineering, which exploits the human factor instead of technical security gaps. In such attacks, the attackers try to mislead users through fake messages, web pages, or even phone calls in order to steal personal and financial information. These attacks are particularly effective because they are based on trust, carelessness, and lack of knowledge, and therefore their prevention is difficult through technical means alone.

A further important category concerns account compromise and the theft of authentication credentials. The use of weak and repeated passwords, together with the absence of additional security mechanisms such as two-factor authentication, makes unauthorized access easier.

Through such attacks, malicious actors can gain access to sensitive data, carry out fraud, or exploit compromised accounts for further malicious actions. This phenomenon highlights the importance of proper credential management and the need to strengthen user knowledge in this area [8].

Phishing attacks remain one of the most frequent and effective forms of cyberattack. They are attacks that rely mainly on deception and manipulation of users with the goal of obtaining sensitive information. Unlike attacks that exploit weaknesses of digital systems, phishing exploits trust, carelessness, and the emotional reaction of users, which makes it particularly dangerous and often difficult to handle. Attackers impersonate legitimate and trustworthy organizations, such as banks or public services, and try to guide victims into actions that may lead to further security violations. Emotional pressure, fear, urgency, and anxiety are used as core manipulation mechanisms.

At the same time, fake news and artificial intelligence form another major challenge. False or misleading news can affect public opinion, influence the way users perceive reality, and undermine trust in media and digital platforms. The use of artificial intelligence has intensified this problem because modern AI systems can now produce text, images, and video that approach human creation to a dangerous degree. Deepfakes in particular create serious risks of misinformation, blackmail, and defamation. However, AI is not only a threat; it is also used as a tool for detecting suspicious information and for analyzing the spread of fake content [9].

3 Digital Learning, Educational Applications, and Gamification

Digital learning is a basic pillar of modern education because it reflects the changes that technology has brought in the way people receive, process, and use knowledge. The growing use of digital devices and services has changed learning to such a degree that it has become more flexible, accessible, and adapted to the needs of different groups of users. Educational applications can therefore act as dynamic tools that support the learning process [10, 11, 12].

Learning through applications is based on software designed for the transmission of knowledge, the development of skills, and the strengthening of active user participation. In contrast to the traditional way, applications allow access to knowledge independent of place and time, facilitating lifelong and self-directed education. Their effectiveness, however, depends strongly on pedagogical design. The simple digitization of content is not enough on its own.

Gamification in learning is a modern educational approach that uses game elements and mechanisms not necessarily for entertainment, but for strengthening participation in learning. Its central characteristic is the active involvement of the user. Through mechanisms such as goals, levels of progress, and rewards, the learner participates actively and receives continuous confirmation of progress. The sense of achievement and gradual progress strengthen motivation, making learning more pleasant and less burdensome.

An important advantage of gamification is learning through trial and error. The user can experiment, make mistakes, and repeat actions without negative real-world consequences, which contributes to better understanding of concepts. Gamification also helps maintain user interest and concentration, especially in digital learning environments, where distraction is always one tap away.

A characteristic example is Duolingo, which uses short lessons, interactive exercises, and reward mechanisms in order to maintain user interest and strengthen the learning process.

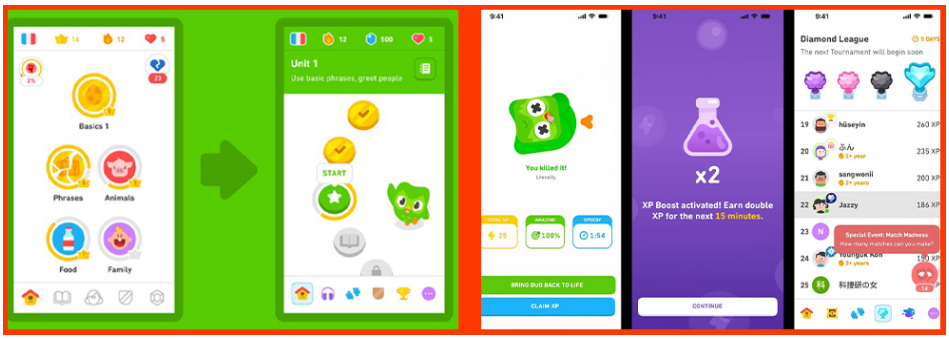


Figure 1: Example of a gamified learning environment in the Duolingo application. Source: Duolingo Blog and Duolingo app screenshot.

These design principles form the basis for Cyberly. The goal is not only to provide information, but also to shape a positive attitude toward learning, prevention, and responsible behavior in the digital world.

4 Cyberly: Idea, Goals, and Target Audience

Cyberly was designed with the goal of prevention and the strengthening of digital safety and critical thinking through education and interactivity. In the modern digital world, where cyberthreats evolve continuously and affect users of every age and level of experience, accessible and effective tools for cybersecurity education are necessary. Cyberly attempts to cover this gap by offering knowledge in a simple, understandable, and user-friendly manner.

The main goal of Cyberly is to raise user awareness regarding both common and modern digital threats, such as phishing attacks, malware, insecure password management, and misinformation on the Internet. Through short educational units and practical examples, the application transforms theoretical knowledge into practical understanding, helping users recognize dangerous situations and adopt safer digital practices.

The application also aims to turn learning into an active process by drawing from digital learning and gamification. Instead of passive reading, users are expected to participate actively through activities, questions, and simulations. In this way, the application strengthens user engagement and helps knowledge remain longer in memory.

An additional important goal of Cyberly is accessibility. The application has been designed so that it addresses a broad audience regardless of age or previous knowledge. The use of simple language and a clear structure allows its use even by users with limited technical experience. The targeted audience includes secondary school students and young adults, adult users with basic digital skills, and educators who may use the application as a supporting tool for information and awareness.

Cyberly was developed with the philosophy of a Minimum Viable Product (MVP), that is, a functional and complete educational application that contains the absolutely necessary characteristics for achieving its basic educational goals. This choice allowed the work to focus on functionality and educational value, avoiding unnecessary complexity and the development of secondary features. The application therefore includes a limited but meaningful set of core functions: short lessons, practical examples, quizzes, simulation, user progress, and supporting tools. At the same time, it leaves open the possibility of future extension.

The educational approach of the application is based on the combination of digital learning, active participation, and gamification. The application avoids passive forms of teaching and encourages participation through interactive activities, questions, and scenarios. The learning structure is based on short and clear units so that learning can become part of the everyday routine of the users without requiring long sessions.

5 UX/UI Design, Onboarding, and Navigation

The user interface design of Cyberly is based on simplicity, clarity, and functionality, with the aim of creating a user-friendly experience. Because the application addresses a broad audience with different levels of digital familiarity, the interface had to remain understandable even for users without previous technical experience. The design followed a mobile-first approach, focusing on use through touch, clear buttons, sufficient distance between interactive elements, and a structure that does not break on different screen sizes.

Special attention was also given to the colors of the application. Colors were selected so as to create a more friendly, modern, and creative environment without strong contrasts that tire the eye. The main interactive elements, such as buttons, lesson cards, and notifications, are visually distinguished from the rest of the content, helping the user quickly recognize actions.

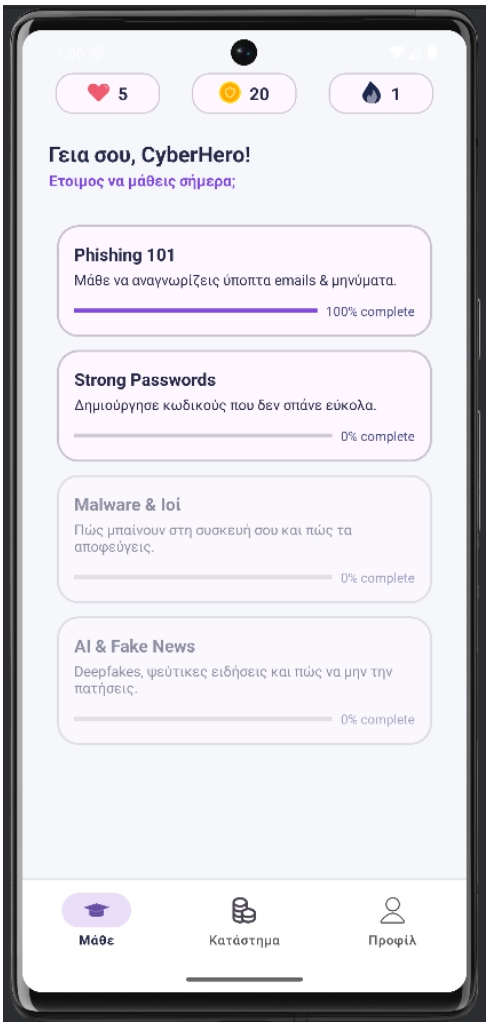


Figure 2: Main screen of the application with lesson cards and basic user interface elements.

The use of cards for presenting learning units was selected in order to make the content more clearly structured. Each unit appears separately, allowing the user to perceive progress and choose the next step with ease. The general aesthetic of the application remains minimalist, with limited decorative elements, so that the user remains focused on the educational content rather than on secondary visual details.

The onboarding process was designed with the purpose of smoothly welcoming the user into the application environment and creating a positive first impression. It does not merely present functions, but aims to activate the user. Through short and understandable messages, the user is informed about the purpose of the application and its learning philosophy. At this stage, the digital mascot Kubo is introduced as a friendly guide of the learning experience. Its role is not purely decorative; it reduces the psychological distance from the subject of cybersecurity, which is often perceived as technical and complicated.

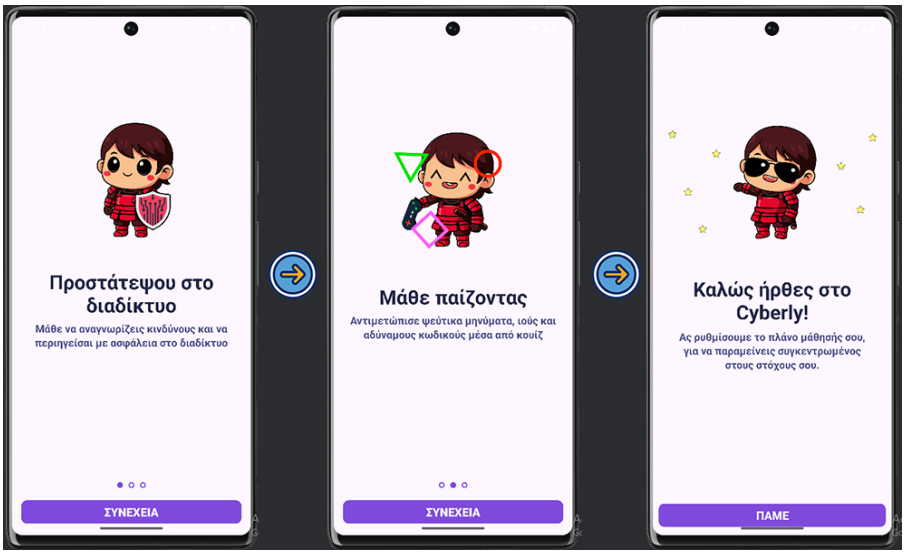


Figure 3: Sequential onboarding screens of the application with the Kubo character.

The application also gives the user the option to define a reminder, promoting consistency in use and helping build study habits. Navigation throughout the application follows the same logic of clarity and simplicity. A fixed bottom navigation menu gives direct access to the basic sections of the application, such as the lessons, the shop, and the user profile. Continue and back buttons remain in stable positions, and each screen gives a clear next action so that the user does not face “dead ends” during navigation.

6 Technologies and Implementation

The development of Cyberly was carried out for the Android platform, a choice based on technical, practical, and strategic criteria. Android is one of the most widespread operating systems for mobile phones on a global scale, which makes it an appropriate environment for the development of educational applications intended for a broad audience. A major reason for this choice is the wide use of Android devices across different price categories, something that increases the accessibility of the application, especially in educational and home environments.

The application was developed in Android Studio, the official integrated development environment for Android applications. Android Studio was used for designing the user interface through XML layouts, allowing the visual design to remain separate from the application logic. The Layout Editor supported the creation of responsive interfaces, while the Android Emulator helped test the application across different Android versions and screen sizes.

The programming language used for the logic of the application is Kotlin. It is a modern language officially supported for Android development, with short syntax and integrated null-safety mechanisms. In Cyberly, Kotlin was used for the implementation of navigation between activities and fragments, for the management of user progress such as lives, experience points, and coins, and for the control of interaction with simulation scenarios. The combination of Android Studio and Kotlin made the development of the application more stable, flexible, and extensible.

The user interface was implemented through XML layouts, separating the presentation level from the logic level. XML also provided flexibility in the structure of screens through the use of layout managers such as `ConstraintLayout` and `LinearLayout`. Particular attention was given to avoiding absolute dimension values and to using guidelines and percentages, so that the application could adapt correctly to different screen sizes.

ViewBinding was adopted for connecting user interface elements to the Kotlin code. This choice avoided the older `findViewById` approach, which requires more manual work and may lead to errors when the IDs or structure of a layout change. In Cyberly, ViewBinding was used across the main screens of the application, including lesson activities, simulation screens, and the user progress environment, contributing to cleaner and safer code.

The application architecture follows a structured approach based on the separation of functions and responsibilities. Navigation is implemented through activities and fragments, while the application structure is divided into packages that separate lesson logic from other functions such as user progress management. The management of lives, coins, XP, and lesson progress is handled through local storage on the device, allowing data persistence each time the application is reopened. This architectural choice aims at a balance between simplicity and extensibility, which fits both the educational goal and the MVP character of the project.

The mascot of the application was not taken from a ready-made template. It was designed through a separate creative process using visual references, character design decisions, and a later animation pipeline. Animation was created in Adobe After Effects, while export to JSON for use in the application was performed so that the animations could be integrated through Lottie. This ensured smooth rendering with relatively low performance overhead.

7 Core Functional Modules and Testing

The educational functionality of Cyberly is primarily organized around the "Learn" section, which combines instructional content, assessment activities, and practical exercises. Each lesson is designed as a self-contained learning unit with a specific educational objective, covering topics such as phishing attacks, malware, password security, and online misinformation. The pedagogical flow of each unit follows three stages: concept introduction, practical examples or scenarios, and knowledge assessment through quizzes.

The learning material is presented through concise educational modules that explain cybersecurity concepts using simple language, visual examples, and scenario-based illustrations. This design aims to make cybersecurity topics accessible to users with limited technical knowledge while maintaining engagement. After studying the theory, users complete short quiz activities that assess their understanding of the presented concepts and encourage active participation in the learning process.

Figure 4 illustrates a representative learning sequence consisting of educational content and a corresponding quiz activity. The combination of instructional content and immediate assessment reinforces knowledge acquisition while preparing learners for practical cybersecurity exercises.

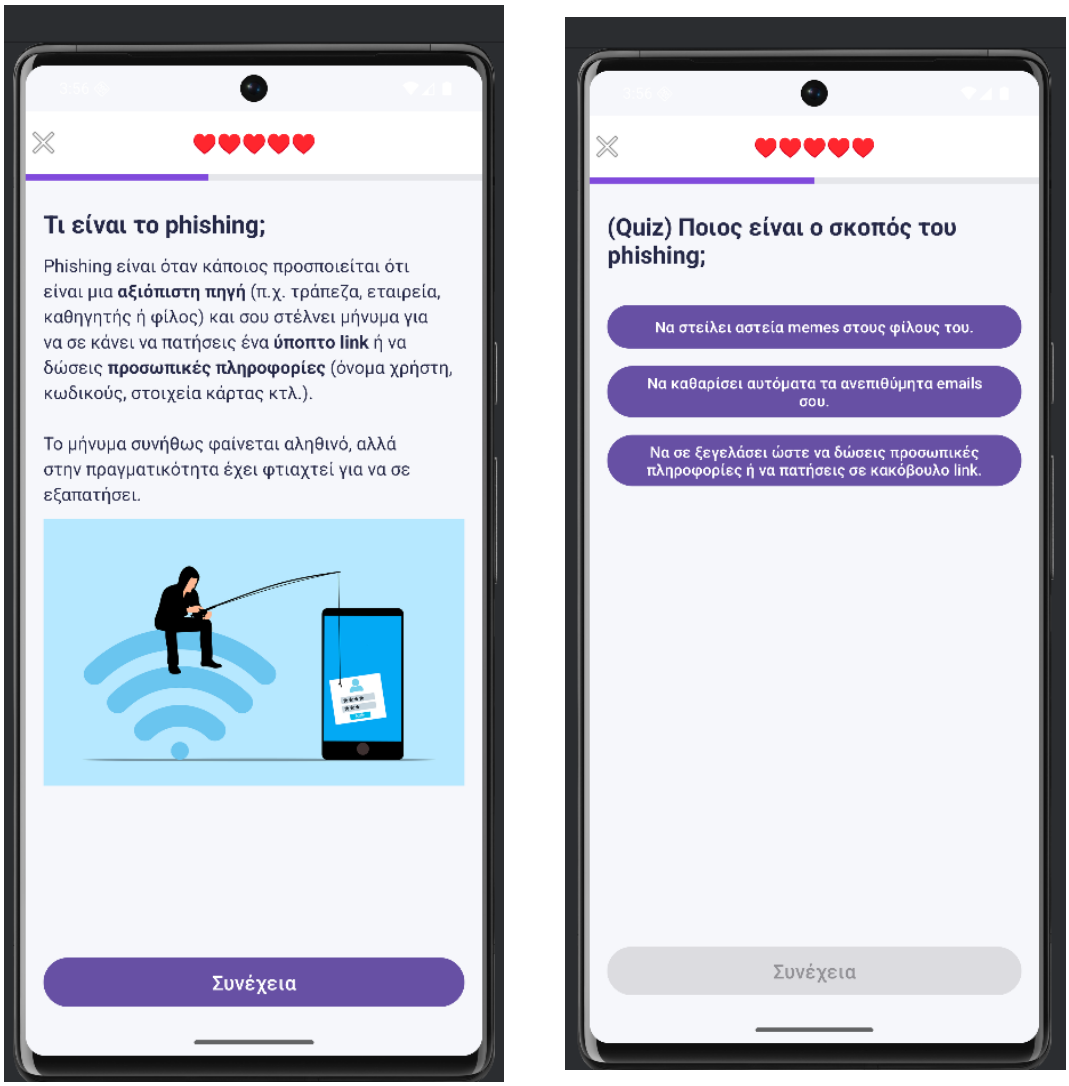


Figure 4: Learning sequence illustrating theory delivery and quiz-based comprehension evaluation.

A central experiential learning component of Cyberly is the interactive phishing simulation. Rather than relying exclusively on theoretical explanations, the application places users in a realistic email environment where they must apply cybersecurity concepts in practice. Before entering the simulation, users receive brief instructions describing the objectives of the activity. They are then asked to inspect incoming messages, identify suspicious indicators, and determine whether an email is legitimate or potentially malicious.

Figure 5 illustrates the complete phishing-awareness workflow, including user guidance, email inspection, and threat identification within a realistic communication environment.

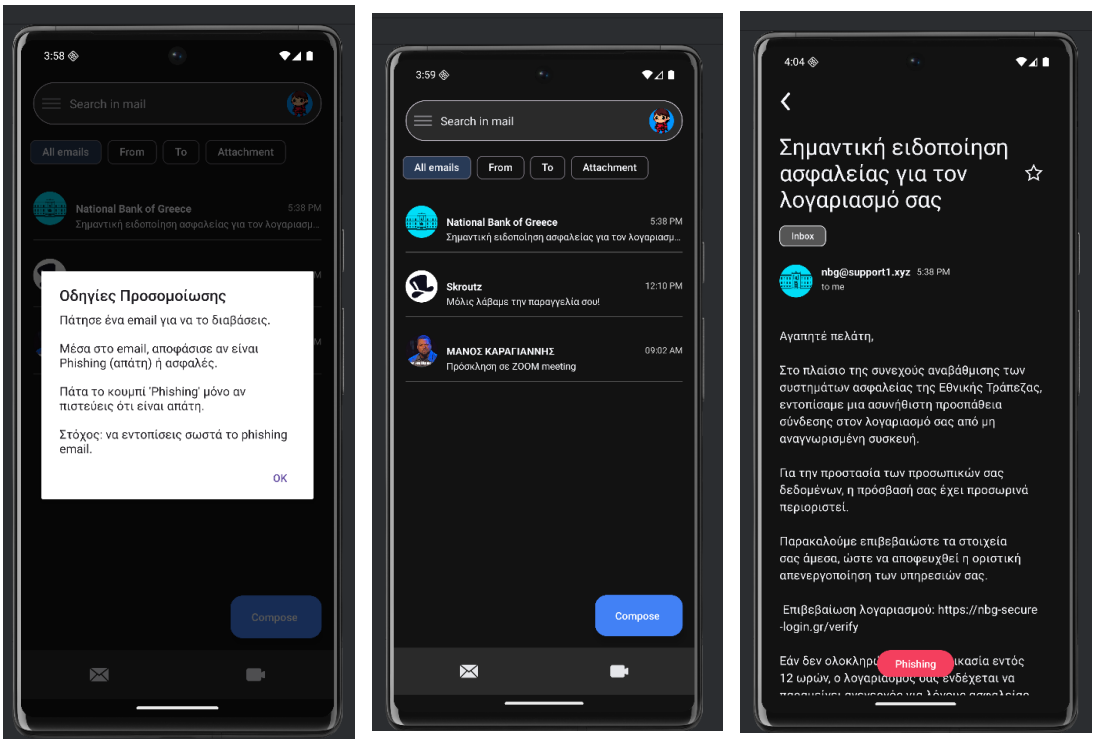


Figure 5: Interactive phishing-awareness scenario demonstrating realistic email inspection and threat identification.

During the simulation, users are required to decide whether a message is safe or suspicious. When an incorrect decision is made, the application does not simply indicate failure; instead, it provides explanatory feedback highlighting the suspicious characteristics of the message. This approach reflects the fact that social engineering attacks primarily exploit human judgment rather than technical vulnerabilities. Consequently, the simulation promotes the development of practical threat-recognition skills and critical decision-making in cybersecurity-related situations.

The progress and reward system constitutes another important component of the gamified learning experience. User progress is represented through experience points (XP), virtual coins, achievement levels, streaks, and available lives. XP is earned through successful completion of lessons, quizzes, and simulation activities, while coins can be used within the application's virtual shop. These mechanisms provide continuous feedback regarding user performance and create short-term and long-term goals that encourage continued engagement.

Figure 6 presents the gamification dashboard, where users can monitor achievements, progression metrics, ranking levels, streaks, and overall learning performance.



Figure 6: User profile dashboard displaying progress metrics, achievements, and engagement indicators.

Supporting functions complement the main educational units. The profile environment allows users to review their overall progress and achievements, while the virtual shop enables the use of coins earned through educational activities. Consistent with the MVP philosophy adopted in this work, these functions were implemented locally without backend dependence or external synchronization.

The control of application functionality was carried out through manual testing. The testing was performed both in the Android Emulator and on a physical Android device in order to confirm the proper adaptation of the interface and the correct operation of the core mechanisms. The tests confirmed that XP remains after restart, coins are updated correctly after purchases, lives do not reset arbitrarily, the streak is maintained according to the application logic, and the reminder is stored correctly. Interface testing also showed that no layout breaks were observed, buttons remained in stable positions, bottom navigation remained functional, lesson cards adapted properly, and animations and the mascot appeared normally and smoothly.

The testing process therefore showed that the application operates stably and according to its design. Although the application does not include automated tests, manual testing covered

the main scenarios of use and confirmed the functional completeness of the MVP.

8 Conclusions

This paper presented the design and development of Cyberly, a gamified tool for learning basic principles of cybersecurity. Through the connection of theory and practical development, the paper presents a complete educational application proposal that responds to a real social need: the strengthening of digital literacy and safer behavior on the Internet.

The development followed the philosophy of a Minimum Viable Product, giving emphasis to basic functionality and stability. The absence of a backend and the limited number of lessons are conscious choices, since they allowed the focus to remain on the educational core and on the quality of the user experience. Despite these limitations, the current version illustrates the practical value of the approach.

Beyond the technical implementation, the application contributes to the broader challenge of strengthening digital education. In an environment where online threats increase day by day and misinformation spreads easily, users need tools that are understandable, practical, and attractive. Cyberly attempts to offer exactly this: a more friendly and accessible way of learning cybersecurity concepts through interactivity, simulation, and reward mechanisms.

Future work will focus on extending the educational content, incorporating personalization features, and assessing the platform's educational effectiveness with real users.

References

[1] ENISA - European Union Agency for Cybersecurity. Cybersecurity overview and threat landscape resources. Available online: https://www.enisa.europa.eu.

[2] F. Giannakas, G. Kambourakis, and S. Gritzalis, "CyberAware: A mobile game-based app for cybersecurity education and awareness," in *2015 International Conference on Interactive Mobile Communication Technologies and Learning (IMCL)*, pp. 54–58, IEEE, 2015.

[3] S. Scholefield and L. A. Shepherd, "Gamification techniques for raising cyber security awareness," in *International Conference on Human-Computer Interaction*, pp. 191–203, Springer, 2019.

[4] M. Malone, Y. Wang, and F. Monrose, "An online gamified learning platform for teaching cybersecurity and more," in *Proceedings of the 22nd Annual Conference on Information Technology Education*, pp. 29–34, 2021.

[5] M. A. Khan, A. Merabet, S. Alkaabi, and H. E. Sayed, "Game-based learning platform to enhance cybersecurity education," *Education and Information Technologies*, vol. 27, no. 4, pp. 5153–5177, 2022, Springer.

[6] ENISA - European Union Agency for Cybersecurity. Phishing and social engineering threats. Available online: https://www.enisa.europa.eu/sites/default/files/publications/ETL2020%20-%20Phishing%20A4.pdf.

[7] ENISA - European Union Agency for Cybersecurity. ENISA Threat Landscape 2025. Available online: https://www.enisa.europa.eu/sites/default/files/2026-01/ENISA%20Threat%20Landscape%202025_v1.2.pdf.

[8] Microsoft Security. What is password protection? Available online: https://www.microsoft.com/en-ie/security/business/security-101/what-is-password-protection.

[9] ACM Digital Library. Research on AI and misinformation. Available online: https://dl.acm.org/doi/fullHtml/10.1145/3584714.3584722.

[10] UNESCO. Digital education - What you need to know. Available online: https://www.unesco.org/en/digital-education/need-know.

[11] European Commission. Role of digital education in promoting digital wellbeing. Available online: https://school-education.ec.europa.eu/en/discover/news/role-digital-education-promoting-young-peoples-digital-wellbeing-and-strengthening-their-resilience.

[12] OECD. Shaping digital education. Available online: https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/07/shaping-digital-education_08b85d69/bac4dc9f-en.pdf.